VICTOR DIAZ GESSNER

victorxdiazg@gmail.com | linkedin.com/in/victordiazgessner | victordiazgessner.github.io/Portfolio

EDUCATION

Northeastern University, Boston, MA — May 2026
Bachelor of Science in Mechanical Engineering — GPA: 3.8/4.0
Achievements/Activities: Dean's List; ASME Treasurer
Certifications: Lean Six Sigma Green Belt

SKILLS

Manufacturing & Prototyping: 3D Printing, Power/Hand tools, DFM, Lab Testing, Fixture Design
CAD: SolidWorks, PTC Creo, Onshape, Siemens NX
Programming & AI: Python, C++ , MATLAB, Minitab, Claude Code, Workflow Automation
Analysis & Testing: ANSYS Fluent, Material Characterization, Instron, RCA
Languages: Native English, Native French, Native Spanish

WORK EXPERIENCE

SharkNinja, Inc, Boston, MA — Jan - Jun 2025
New Product Development Engineering Co-op
Engineered next generation wet floorcare system through CAD-driven design, fluid dynamics calculations, and validation testing, coordinating across NPD, PD, manufacturing design and EE teams.

- Co-invented a squeegee architecture for next generation wet/dry vacuums, submitted for patent reivew
- Built Python manifold tool, reducing physical iteration from 12 prototype builds to 3 targeted variants
- Created Creo prototype assemblies, translating performance requirements into mechanical design changes
- Tested prototypes against flow-rate, pickup, and usability targets to guide design tradeoffs
- Worked with product design, manufacturing design, and EE teams on feasibility, packaging, and DFM
- Presented findings through design reviews, prototype demos, and engineering-leadership meetings

DAPS Research Laboratory, Boston, MA — Aug – Dec 2024
Undergraduate Research Associate
Performed material characterization research for aerospace/nuclear shielding, investigating quantum-scale interactions in UV-curable ceramics through nuclear reactor experiments for Raytheon.

- Optimized ceramic material compositions through statistical analysis of reactor neutron flux data, improving shielding performance by 43%
- Designed and fabricated iterative ceramic prototypes using additive manufacturing
- Modeled mechanical properties using Jacobian-based analytical methods
- Validated microstructures & material performance through Scanning Electron Microscopy (SEM) imaging

Johnson & Johnson MedTech, Danvers, MA — Jan - Jun 2024
Production Engineering Co-op
Optimized high-precision medical-device manufacturing through data-driven process improvements, statistical analysis, and Lean Six Sigma methodologies across multiple production lines.

- Increased yield 23% through Lean analysis, defect tracking, SPC, and operator process improvements
- Supported 6x weekly output increase through line-layout analysis, operator coordination, and 156 production fixtures
- Prevented $1.3M+ in losses through Instron testing, FMEA workflows, Gage R&R, and CAPA support

ENTREPRENEURSHIP AND LEADERSHIP

Engineering Forward, Inc. Boston, MA — Jan 2025-Present
Co-Founder & President
Founded and incorporated a peer mentorship nonprofit that connects students with graduating seniors or recent graduates who have firsthand experience navigating the same academic and professional challenges.

- Scaled program to 58 mentees and 10 mentors with experience in companies like RTX, J&J, and SharkNinja
- Incorporated as a Massachusetts nonprofit pursuing 501(c)(3) status, developed bylaws & a board of directors